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RESEARCH

Enhanced selective hyper chaotic encryption using deep neural network for hyperspectral images

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Abstract
Hyperspectral images are crucial for remote sensing, military applications, medical imaging, and satellite navigation systems. Maintaining the confidentiality, privacy, security, and intellectual property rights of the HSIs is essential. However, generalized traditional encryption methods are less secure and robust due to larger spectral bands, high variability in spectral bands, redundancy in spectral bands, and higher encryption time. This article presents selective hyperspectral image encryption based on a deep learning-based hyperchaotic image encryption algorithm that improves the security and robustness of the hyperspectral images. Here, a deep neural network is utilized to enhance the randomness of the hyperchaotic sequence key and improve encryption. The performance of the deep learning-based hyperchaotic image encryption is estimated on the PAVIA, SALINAS, Indian Pines, Kenndy Space Center, and Cuprite datasets based on Mean Square Error, Peak signal-to-noise ratio, Structural Similarity Index metrics, Correlation Coefficient, Number of Pixel Change Rate, Unified Average Changing Intensity and entropy. The proposed method offers the MSE in the range of 1.946 to 2.007, PSNR in the range of 45.10 dB to 46.31 dB, SSIM in the range of 0.953 to 0.982, CC in the range of 0.02 to 0.015, NPCR in the range of 98.34, UACI in the range of 27.59 for five sample hyperspectral images. The proposed scheme provides improved security and robustness against distinct encryption attacks such as Brute Force Attack and chosen-plaintext-attack. It offers superior entropy of 7.85 for brute force attack and 7.83 for the chosen-plaintext attack.

Keywords Hyperspectral image encryption · Hyperchaotic image encryption · Deep neural network · Remote sensing

Introduction

Hyper-spectral imaging is wisely utilized in many applications such as military application, geographical monitoring, vegetation detection, mineral detection, and remote sensing applications. Hyperspectral images (HSIs) have multiple spectral bands that are capable of distinguishing different objects uniquely. However, the authentication, privacy, and security of the HSIs are challenging because of their intricate structure. Encryption helps to preserve the authenticity and privacy of images. Security techniques for traditional images, such as watermarking, transform-domain techniques (e.g., DCT, DWT), cryptographic methods (AES, RSA), chaotic encryption, and deep learning-based methods, can be adjusted for HSI to account for its larger size and multidimensional structure. Cryptographic techniques provide strong data protection, but because of their sensitivity to initial conditions and ability to conceal both spectral and spatial data, spectral band permutation and chaotic encryption are particularly effective for HSI.

Image encryption is the method of converting images from readable to non-readable formats. Encryption is generally categorized into asymmetric and symmetric encryption (SaberiKamarposhti et al. 2024). In asymmetric method, for encryption public key is utilized and for decryption the private key is utilized. RSA is a significant example of

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asymmetric encryption (Kolivand et al. 2024). Asymmetric encryption is more secure but slower and cannot be utilized for larger images. Symmetric encryption uses the same key for encryption and decryption. Data encryption standard and Advanced encryption standard are popular symmetric encryption schemes. Symmetric encryption is efficient and faster, which makes it more popular for image encryption (Alghamdi and Munir 2024).

Encryption protects the confidential information of the images, prevents unauthorized access, detects and avoids unauthorized tampering of the images, secures the content, prevents unauthorized access, and preserves the user's intellectual property rights (Wen et al. 2024; Toktas et al. 2024). The conventional encryption algorithms utilized for natural image encryption are incompatible with hyperspectral or multi-spectral image encryption because of high-dimensional data, complex bands, high variability in spectral bands, redundancy in spectral bands, and high processing time due to large spectral bands. HSI image encryption is crucial because HSIs are utilized in remote sensing, medical imaging, and military surveillance (Zhang and Hu (2024)). Therefore, it is essential to maintain the confidentiality, privacy, security, integrity, and intellectual property rights of HSIs. This helps to maintain the authenticity and secured communication of HSI data (Audhkhasi and Povinelli 2021). Hyperchaotic image encryption (HCE) algorithms use hyperchaotic systems for image encryption. Hyperchaotic systems use multiple Lyapunov exponents to exhibit a complex and high-dimensional chaotic nature. The complexity and unpredictable nature of HIE makes it more robust for secure encryption. HIE provides strong sensitivity, enhanced security, and excellent diffusion (Srujana et al. 2020).

The generalized process of HIE consists of initializing a initial conditions and chaotic system. It also generates the pseudorandom sequence encryption key. In the next stage, permutation and diffusion of pixels are utilized to improve the randomness of data. Finally, the image is encrypted using a hyper-spectral key. Zhang and Yan (2021); Lin et al. (2021). However, the system's effectiveness is limited because of the increased complexity of many hyperspectral bands. The encryption performance depends upon the randomness of the key, where poor randomness results in poor encryption quality and is prone to different encryption attacks.

This paper presents the deep learning-based hyper-chaotic selective hyperspectral image encryption to enhance the robustness and resilience against attacks. The results of the DLHCE is assessed on the PAVIA, SALINAS, IndianPines, KSC, and Cuprite datasets based on Mean Square Error (MSE), Structural Similarity Index metrics (SSIM), Peak signal-to-noise ratio (PSNR), Correlation Coefficient (CC), Number of Pixel Change Rate (NPCR), Unified Average Changing Intensity (UACI) and entropy.

An overview of the paper's main contributions is provided below:

- In the proposed method, hyperchaotic image encryption is carried for hyper spectral images for selective bands based on entropy and variance, reducing the computational overhead.
- Hyperchaotic key enforcement using deep neural networks (DNN) to enhance the robustness against different attacks such as Brute Force Attack (BFA), man-in-middle (MIM) attack and chosen-plaintext-attack (CPA) to boost the efficiency of the encryption method.
- Analysis of the different encryption attacks on the DLHCE scheme to validate the robustness of the encryption method.

The article is arranged as follows: Section"Related work" provides the related work regarding hyperspectral image encryption. Section"Proposed selective band encryption using enhanced hyperchaotic map using deep neural network" offers the proposed methodology in detail. Section"Experimental results and discussions" delivers the simulation results for the encryption and attacks on the encryption. Lastly, Section"Conclusion" depicts the findings and future directions of the work.

Related work

This section offers an overview of the different encryption techniques that have previously been proposed. Various researchers have focused on HSI encryption in the last decade because of its complexity, large band variance, and redundancy in spectral bands. The HSIs have shown a massive demand for encryption and cryptography because of ethical issues, security, and intellectual property rights. Sharma et al. (2024) presented a chaos encryption algorithm to protect the hyperspectral image (HSI) information. Self Adaptive Bald Eagle Search algorithm is used to optimized the parameters of the chaos technique. It uses circular shift in a multi-level chaos map-based encryption algorithm. It has shown robustness for different attacks on medical hyperspectral images, such as cropping, differential noise, and statistical attacks. However, the algorithm's effectiveness is limited because of the poor structural content of the images.

Xiao et al. (2024) suggested a 4D Lorenz hyperchaotic system based on brushing diffusion and 3D zig-zag transformation. It used a 4D Lorenz hyperchaotic system to create the scrambling random sequence followed by brushing diffusion to boost the avalanche effect. The 3D zig-zag transformation minimizes the inter-band and intra-band correlation. The suggested scheme provided better randomness in the cipher text, a large key space, excellent visual reconstruction quality, high

key sensitivity, and a quick process. The band redundancy in the HSI leads to lower efficiency for the data hiding in the HSIs than natural images.

Fan et al. (2022) investigated a novel Reversible data hiding method that uses pixels from the adjacent bands while data hiding. The RDH-based encryption's visual quality is limited due to noise and hiding multiple band data. Yang et al. (2022) explored the triangular association encryption algorithm, and Arnold transform, which considers the diffusion of the HSI pixels within the band. The image signature is used for the HSI restoration to enhance the robustness of the encryption process. Hangyu and Zhi (2022) presented the compression using Walsh transform and encryption using Hadamard permutation for HSI-based secure fingerprint recognition. However, the visual quality and structural quality of the HSI content.Numerous scholars have innovatively adapted and improved existing chaotic schemes to increase chaos's applicability in image encryption algorithms.

Erkan et al. (2022) suggested the modified chaotic map considering the Pi and Euler numbers for encryption. This scheme provided better complexity, ergodicity, and randomness randomness of the hyper-chaotic system. It used the bit reversal diffusion technique for image encryption but took longer. Further, Tian et al. (2024) used curve cryptosystem and Fibonacci sine exponential map for HCE. It utilized the 3-D scrambled diffusion technique to enhance security and efficiency. However, the scheme results are challenging due to the intricate multi-band structure of images. Traditional systems utilize absolute moment blocks truncation coding to compress HSI, which suffers from inferior irreversibility and perceptibility. Zhang et al. (2024) explored bitmap prediction for encryption the message to preserve the structural quality and information of the image. It provides better perceptibility and irreversibility for recovering the original information.

Bodke and Chaudhari (2024) suggested the hybrid combination of forensic optimization and Archimedes optimization for the selection of the salient spectral band for embedding the watermark. The discrete wavelet transform-based watermarking scheme has shown the robustness and security of different geometrical and noise attacks on hyperspectral images. Chen and Lv (2024) invented a 2D sine-cosine-logistic coupling chaotic system for image encryption. It minimizes the lower complexity and discontinuities in the random sequence utilized for the encryption. It offered superior security than existing chaotic algorithms and higher information entropy. However, the scheme is challenging for multi-spectral images because of the complexities and disparities in the spectral bands. Alsubaei et al. (2023) presented block scrambling for hyperspectral image encryption to preserve the information and maintain the multi-band images' overall perceptual, visual, and structural quality. In block scrambling, the original image is split into non-overlapping blocks before the encryption process, which shows robustness against the attacks. Alexan et al. (2023) proposed a multi-layer encryption scheme to enhance the security based on a fractional-order Chen hyper-chaotic map. It has shown robustness against brute force and plaintext attacks based on entropy and visual and statistical measures. Gao et al. (2023) presented HCE based on a discrete resistance-coupled sinusoidal map. It further uses DNA mutation to improve the complexity of the HCE system.

Chen et al. (2023) collaborate with a cubic map and an improved cosine map for selective encryption of medical images. The algorithm is challenging due to higher encryption time that makes it unsuitable for real time encryption. Further, Huang et al. (2023) explored the sinusoidal cascade of 1-D chaotic mapping, which is identical in nature to building the chaotic system. However, the intricacy of the system makes it difficult for the HCE. Wang et al. (2024) developed an encryption scheme for images having irregular size content. It utilized a novel logistic sign chaos system, which uses a sinusoidal map for perturbing control variables and a logistic map for adding state delay. However, the intricate encryption process and poor time efficiency make practical deployment for complex HSI encryption infeasible.

Tang et al. (2021) suggested a discrete Fourier transforms and logistic-exponent-sine map are used to suggest an encryption solution for multiple color images. Each color image is first represented as a trinion matrix, and then block-wise discrete trinion Fourier transforms are applied. The real and imaginary components of the altered findings are then combined to create the real matrix. A second real matrix that utilizes only half of the spectrum based on conjugate symmetry is then produced by applying a two-dimensional discrete Fourier transform. Shao et al. (2020) presents a novel encryption algorithm that securely transmits many images at once by combining quaternion gyrator transform with equal modulus decomposition.To improve the encryption security, the algorithm employs equal modulus decomposition and a chaotic random phase mask.

Limitation of previous work

A number of techniques have been used to encrypt hyperspectral images. The following is a summary of the prior work's limitations:

- Encrypting a entire hyperspectral image requires more computational resources, leading to slower encryption and decryption times and higher CPU and memory usage, impacting system performance, especially on resource-limited platforms.
- Whole-image encryption requires sophisticated key management, increasing complexity. It increases the risk of key mismanagement and reduces granularity in encryption policy. Selective encryption allows different security

levels, while whole-image encryption uses the same key for all data.

- The encryption scheme's effectiveness is hugely reliant on the encryption key, its randomness, and the initial parameters of the encryption algorithm. Traditional methods are complex in nature because of redundancy and high variability in spectral bands. The effectiveness of the existing scheme is challenging because of different encryption attacks.
- The outcomes of the chaotic encryption algorithms are mostly dependent on the precision of the initial conditions and randomness of the key. The chaotic encryption algorithms are prone to different attacks, such as chosen and known plain-text attacks. It provides lower resistance to distortions and noise in the images.

Therefore, this article provides a Selective band encryption using enhanced hyperchaotic map using Deep Neural Network.

Proposed selective band encryption using enhanced hyperchaotic map using deep neural network

The proposed DLHCE consists of four major steps: spectral band selection, image encryption using the HCE algorithm, randomness optimization of the key sequence using lightweight DNN, and image decryption. The effectiveness of the algorithm is assessed based on different encryption attacks such as brute force attack (BFA), man-in-the-middle (MIM), and chosen-plaintext attacks (CPA). The performance of the encryption scheme is evaluated based on PSNR, MSE and entropy.

Architecture of proposed framework

In this work, selective band encryption using enhanced hyper chaotic method using deep neural network is proposed. The proposed work is carried out in two phases:encryption and decryption of the image. In the first phase, the input will be hyper spectral image with multiple bands, then band selection process will be carried out based on maximum entropy and variance to protect the selective bands and minimize the computational complexity. Then, to encrypt the selected bands keys are generated using hyperchaotic method. Furthermore, to increase the security and provide robustness against encryption attacks, hyperchaotic keys are enhanced using deep neural network. Enhanced hyperchaotic keys are used for encrypting the selected bands which have more randomness. On the other hand, decrytion process will be carried out to obtain the original images. Figure 1 provides the flow diagram of the DLHCE algorithm.

Hyperspectral band selection

The proposed method uses HIS images as the input. The HISs have multiple bands with different degrees of variability and information. Processing all bands increases the complexity and processing time of the encryption scheme, which limits the performance of the real-time implementation of the encryption scheme. We have considered selective encryption that encrypts the spectral bands with high entropy (EN) and standard deviation (SD).The EN predicts the unpredictability and randomness of the pixel value in the HSIs, which has a value between 0 to 8 because the image consists of 256 levels of intensity (8 bits). EN offers the information associated with the spectral bands as given in Eq. 1. Here, p(i) indicates

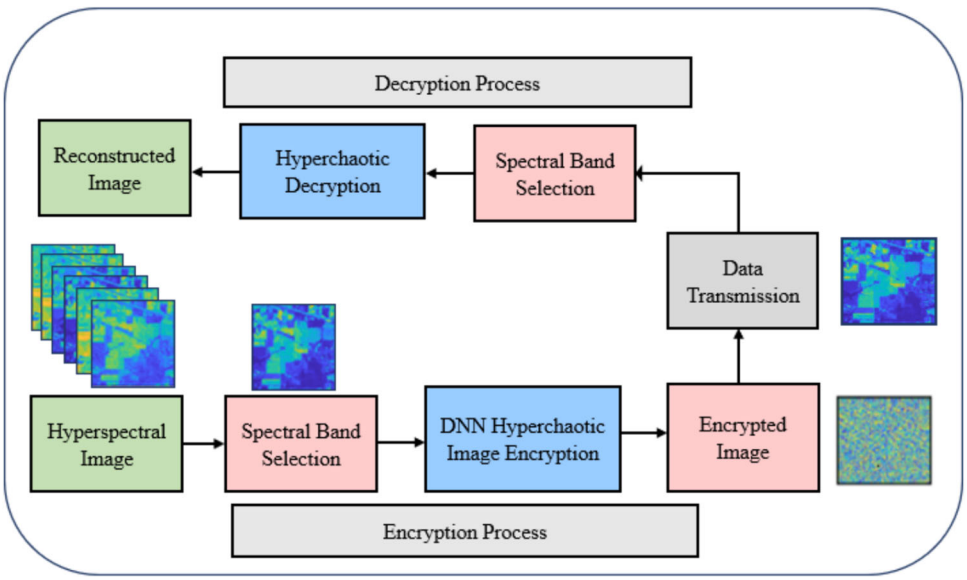


Fig. 1 Illustration of Proposed Deep Learning-based Hyperchaotic Image Encryption and Decryption process

the probability of each pixel in HIS.

$$EN = -\sum_{i=0}^{255} p(i)\log_2(p(i)) \tag{1}$$

The SD offers a disparity in the intensity and contrast of the pixels around the mean intensity. The SD is computed using Eq. 2. Here, PN represents the total pixel in the HIS band, $h(i)$ is a pixel in the HIS band, and μ denotes the mean of the HIS band as given in Eq. 3.

$$SD = \sqrt{\frac{1}{PN}\sum_{i=1}^{PN}(h(i)-\mu)^2} \tag{2}$$

$$\mu = \frac{1}{PN}\sum_{i=1}^{PN} h(i) \tag{3}$$

The spectral bands of the HSIs are selected using Eq. 4, where α is the weight factor for entropy, which is considered 0.6. We have performed the experimentation for the different values of the α starting from 0.1 to 0.9 where it results in higher PSNR for α=0.6. This shows that spectral band information is given higher weightage compared with the overall pixel distribution.

$$HSI_{\text{band}} = \alpha * EN + (1-\alpha) * SD \tag{4}$$

The spectral band with higher entropy consists of diverse, non-distinguishable patterns, and random pixels that are harder to predict. This randomness is best suited for encryption as it prevents different attackers from exploiting patterns. Higher variance ensures variability in the distribution of pixels, which reduces the risk of pattern or redundancy in the cipher data. The higher value of entropy and variance makes the spectral band's encryption more robust and secure, and avoids data redundancy. However, lower values of entropy and variances are prone to different attacks and provide less security and robustness due to redundancy and discernible patterns.

Hyperchaotic image encryption

For the encryption of selected spectral bands, the hyperchaotic encryption method is used. The hyperchaotic encryption stepwise process is shown in Fig. 2. The hyperchaotic sequence key is enhanced using DNN to boost the randomization of the encryption key. A hyperchaotic sequence key is a series of random values generated from a hyperchaotic system, which has multiple positive Lyapunov exponents and displays complex, unpredictable behavior. It is derived from the system's state variables and enhances encryption security by ensuring high sensitivity to initial conditions and randomness. The key is often mapped to an 8-bit range for operations such as pixel permutation and diffusion in image encryption. The steps in the hyperchaotic encryption algorithm are given as follows.

- Step 1: Selection of the hyper-chaotic system.
  The hyper-chaotic system is defined by the Eq. 5. Here, x, y, z, and w are state variables; a, b, c, and d are system parameters such that a=10, b=28, c=8/3, d=1. The differential variables $\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}, \frac{dw}{dt}$ denote the rate of change of state variables with respect to time.

$$\begin{aligned} \frac{dx}{dt} &= a(y-x) \\ \frac{dy}{dt} &= cx - y + wz \\ \frac{dz}{dt} &= -bz + xy \\ \frac{dw}{dt} &= d(x-w) \end{aligned} \tag{5}$$

- Step 2: Selection of initial conditions for state variables.
  The initial conditions for the hyperchaotic system are considered as given below in the following Eq. 6. Here,

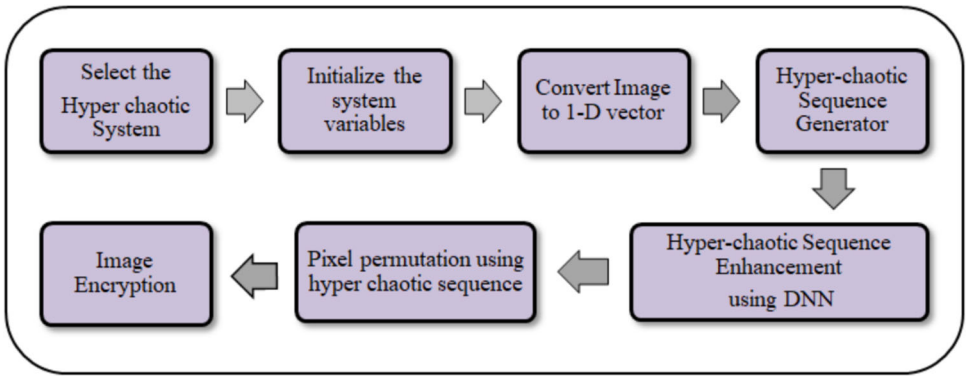


Fig. 2 Illustration of Hyperchaotic Encryption process

x(0),y(0),z(0) and w(0) symbolizes the initial conditions of state variables x, y, z and w respectively.

$$\begin{aligned} x(0) &= 1 \\ y(0) &= 1 \\ z(0) &= 1 \\ w(0) &= 1 \end{aligned} \tag{6}$$

- Step 3: Convert image to 1-D vector In this stage, the M×N grayscale image is converted to a 1D array (P), which is represented by Eq. 7, where M and N denotes the rows and columns of image.

$$P = \left\{P_{ij}\right\} \quad , 1 < i < M, 1 < j < N \tag{7}$$

- Step 4: Chaotic Sequence Generator
The chaotic sequence is generated by solving the Eq. 8, where s_k indicates the chaotic sequence of state variables at time t_k, and $x(t_k)$, $y(t_k)$, $z(t_k)$, and $w(t_k)$ are the state variables at time t_k.

$$\begin{aligned} S &= \{s_k\} \\ \{s_k\} &= \lfloor (x\,(t_k) + y\,(t_k) + \mathrm{z}\,(t_k) + \mathrm{w}\,(t_k)) \bmod 256 \rfloor \end{aligned} \tag{8}$$

- Step 5: Enhance the randomness of the Chaotic Sequence using DNN.
- Step 6: Pixel permutation using hyper-chaotic sequence. The pixels in the 1D image array are permuted according to the chaotic sequence. If the sequence is S=s1,s2,…, S_{mn}, then the permutation can be described using Eq. 9. Here, P' denotes the permuted pixel value, P'_k denotes the k-th permuted pixel in array P', and $P_s[k]$ indicates the pixel in the original array P at the position given by the sequence $s[k]$.

$$P' = \left\{P'_k\right\}, P'_k = \mathrm{Ps[k]} \tag{9}$$

- Step 7: Image Encryption using Pixel diffusion Each pixel value in the permuted array is modified by a diffusion process, typically involving XOR operations with the chaotic sequence and previous pixel values,as shown in Eq. 10, where d_k indicates the k-th pixel in the diffused array, d_{k-1} denotes the $(k-1)$ pixel in the diffused array, and d_0 stands for the initial position of the diffusion process.

$$d_k = \left(P'_k \oplus s_k\right) + d_{k-1}, \quad d_0 = s_0 \tag{10}$$

- Step 8: Reconstruct the original image to M×N Pixel to get the encrypted image HSI_{enc}. The hyperchaotic sequence consist of values between 0 and 256 because of 8 bit unsigned integer format of the image.
- Step 9: During the decryption stage, the spectral bands are selected from the decrypted HSI HSI_{enc}. Then the selected bands are converted to 1-D vector and regenerated using chaotic sequence used during encryption. Then reverse pixel diffusion and permutation is applied to decrypt the sequence. The 1-D sequence is converted back to 2-D format to get decrypted image.

Deep neural network for hyperchaotic image encryption

The DNN enhances the randomness of the encryption key sequence generated by the traditional hyper-chaotic encryption algorithm. The DNN accepts the key generated by the HCE algorithm and provides an enhanced key with higher randomness. The randomness of the key sequence is evaluated using the serial correlation test (SCT). The SCT provides the correlation of the key with the lagged (adjacent values) version of the key. The objective function for selecting the encryption key (s) is given by Eq. 11.

$$\mathrm{SCT} = \frac{\sum_{i=1}^{PN-1}((s_i - \bar{s})(s_{i+1} - \bar{s}))}{\sum_{i=1}^{PN}\left((s_i - \bar{s})^2\right)} \tag{11}$$

The suggested DNN consists of three fully connected layers (FC), three rectified linear unit layers (ReLU), three batch normalization layers (BN), and a regression layer as shown in Fig. 3.

DNN consist of three FCL with 20, 40 and 60 hidden layers. The increasing hidden layer improves the hierarchical feature representation of the input sequence. The FC layer output for the gth layer is given by Eq. 12. Here, $h^{(g)}$ denotes the output of the gth hidden layer, s is the input to the FC layer, $b(g)$ is the bias value for the gth hidden layer, and w_g stands for the weights of the gth hidden layer. The FC layers offer interconnectivity and correlation in the input encryption sequence. Multiple FC layers enhance the hierarchical representation of the sequence.

$$\begin{aligned} h &= \rho\left(W^{(1)} \cdot s + b^{(1)}\right) \\ h^l &= \rho\left(W^{(l)} \cdot h^{(l-1)} + b^{(l)}\right) \end{aligned} \tag{12}$$

The ReLU layer boosts the non-linearity of the features by choosing the positive attributes and replacing negative neurons with zero, as given in Eq. 13. It boosts training performance and helps lessen the gradient vanishing problem.

$$R = \max(0, h) \tag{13}$$

The BN layer normalizes the ReLU layer output by considering the mean (mr), standard deviation (sr), scale parameter

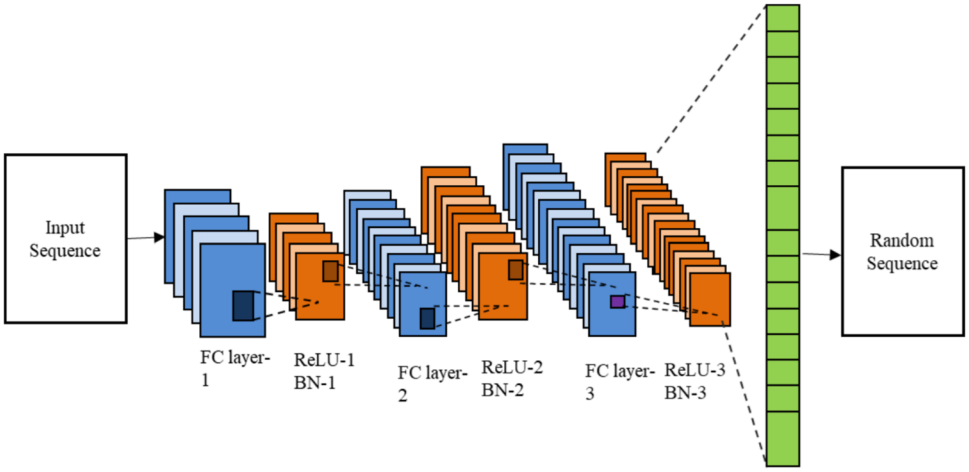


Fig. 3 Framework of Proposed DNN for Key Enhancement

(α), and offset parameter (β) for a batch size of 32, as given in Eq. 14.

$$BN = \alpha \frac{R - mr}{sr} + \beta \tag{14}$$

The regression layer predicts the randomized sequence at the output layer using Eq. 15.

$$\hat{s} = w^{(g+1)} h^{(g)} + b^{(g+1)} \tag{15}$$

The original and enhanced hyperchaotic key is represented using Fig. 4. The DNN minimizes the dependencies in the successive samples and helps to randomness of the key. The original sequence in Fig. 4 provides an SCT of 0.875 (low randomness), and the enhanced sequence provides an SCT of 0.034 (high degree of randomness).

The selective band encryption using enhanced hyper chaotic method using deep neural network is illustrated using HIS encryption Algorithm 1 and HIS decryption Algorithm 2.

Algorithm 1 Proposed DLHCE algorithm for HIS encryption.

Require: Hyper-spectral Image(HSI)

1: **Process:**

1. Selection of spectral band HSI_{band} using Eq. 4.
2. Selection of the hyper-chaotic system using Eqs. 5-8.
3. Selection of initial conditions for state variables using Eqs. 9-12.
4. Convert image to 1-D vector using Eq. 13.
5. Generate chaotic random sequence using Eqs. 14-15.
6. Enhancement of randomness of hyper-chaotic random sequence using DLL.
7. Pixel permutation using hyper-chaotic sequence Eq. 16.
8. HSI encryption using pixel diffusion Eq. 17.

Ensure: Encrypted HSI bands(HSI_{enc}).

Algorithm 2 Proposed DLHCE algorithm for HIS decryption.

Require: Encrypted Hyper-spectral Image (HSI_{enc})

1: **Process:**

1. Selection of encrypted spectral bands.
2. Convert the bands in to 1-D vector.
3. Regenerate chaotic sequence.
4. Apply reverse pixel diffusion.
5. Apply reverse pixel permutation.
6. Reshape 1-D vector to its original 2-D format.

Ensure: Decrypted HSI bands(HSI_{dec}).

Sample example

- **Step 1:** Consider the selected HSI bands of 4×4 pixels based on entropy and variance as given below:

$$\text{im} = \begin{bmatrix} 120 & 150 & 180 & 200 \\ 130 & 160 & 190 & 210 \\ 140 & 170 & 200 & 220 \\ 150 & 180 & 210 & 230 \end{bmatrix}$$

- **Step 2:** Convert the image into a 1-D vector as:

$P = [120, 130, 140, 150, 150, 160, 170, 180, 180, 190, 200, 200, 210, 220, 230]$

- **Step 3:** Generate the hyperchaotic key using Eq. 8:

$S = [100, 150, 200, 50, 180, 220, 130, 170, 150, 200, 180, 170, 220, 210, 240, 130]$

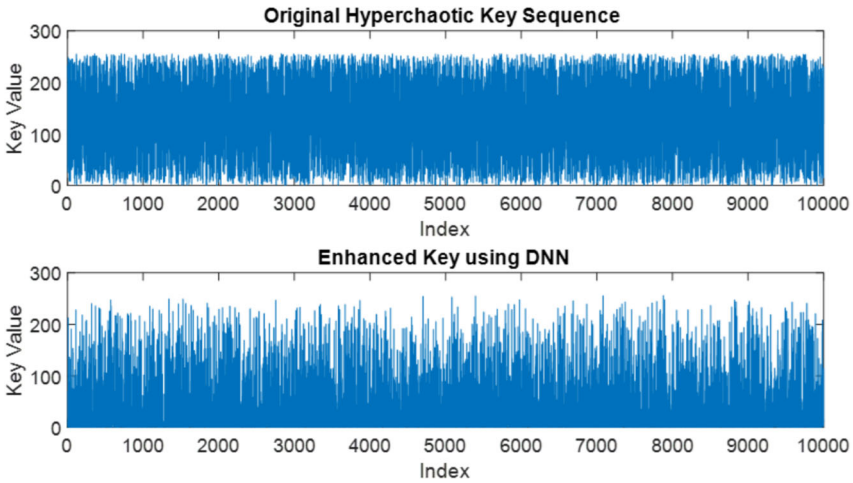


Fig. 4 Original key and enhanced key

- **Step 4:** Enhance the randomness of the hyperchaotic encryption sequence using DNN:

  $S_{\text{enhanced}} = [120, 170, 220, 60, 190, 230, 140, 180, 160, 210, 190, 180, 240, 220, 250, 150]$

- **Step 5:** Using the permutation rule $P' = \{P'_k\}$, $P'_k = P_s[k]$, the image pixels are rearranged according to the values in the enhanced chaotic sequence:

  $P' = [150, 180, 200, 220, 170, 210, 160, 240, 140, 180, 170, 190, 230, 240, 220, 250]$

- **Step 6:** Perform the encryption using pixel diffusion with the equation:

  $$d_k = (P'_k \oplus s_k) + d_{k-1}, \quad d_0 = s_0$$

  Initial Condition $d_0 = s_0 = 120$

- Pixel 1: $d_1 = (P'_1 \oplus s_1) + d_0 = (150 \oplus 170) + 120 = 20 + 120 = 140$
- Pixel 2: $d_2 = (P'_2 \oplus s_2) + d_1 = (180 \oplus 220) + 140 = 100 + 140 = 240$
- Pixel 3: $d_3 = (P'_3 \oplus s_3) + d_2 = (200 \oplus 60) + 240 = 140 + 240 = 380 \pmod{256} \Rightarrow 124)$
- Pixel 4: $d_4 = (P'_4 \oplus s_4) + d_3 = (220 \oplus 190) + 124 = 90 + 124 = 214$
- Pixel 5: $d_5 = (P'_5 \oplus s_5) + d_4 = (170 \oplus 230) + 214 = 60 + 214 = 274 \pmod{256} \Rightarrow 18)$
- Pixel 6: $d_6 = (P'_6 \oplus s_6) + d_5 = (210 \oplus 230) + 18 = 80 + 18 = 98$
- Pixel 7: $d_7 = (P'_7 \oplus s_7) + d_6 = (160 \oplus 140) + 98 = 20 + 98 = 118$
- Pixel 8: $d_8 = (P'_8 \oplus s_8) + d_7 = (240 \oplus 180) + 118 = 60 + 118 = 178$
- Pixel 9: $d_9 = (P'_9 \oplus s_9) + d_8 = (140 \oplus 160) + 178 = 20 + 178 = 198$

$E = [140, 240, 124, 214, 18, 98, 118, 178, 198, 228, 32, 42, 52, 112, 142, 242]$

- Pixel 10: $d_{10} = (P'_{10} \oplus s_{10}) + d_9 = (180 \oplus 210) + 198 = 30 + 198 = 228$
- Pixel 11: $d_{11} = (P'_{11} \oplus s_{11}) + d_{10} = (170 \oplus 190) + 228 = 60 + 228 = 288 \pmod{256} \Rightarrow 32)$
- Pixel 12: $d_{12} = (P'_{12} \oplus s_{12}) + d_{11} = (190 \oplus 180) + 32 = 10 + 32 = 42$
- Pixel 13: $d_{13} = (P'_{13} \oplus s_{13}) + d_{12} = (230 \oplus 240) + 42 = 10 + 42 = 52$
- Pixel 14: $d_{14} = (P'_{14} \oplus s_{14}) + d_{13} = (240 \oplus 220) + 52 = 60 + 52 = 112$
- Pixel 15: $d_{15} = (P'_{15} \oplus s_{15}) + d_{14} = (220 \oplus 250) + 112 = 30 + 112 = 142$
- Pixel 16: $d_{16} = (P'_{16} \oplus s_{16}) + d_{15} = (250 \oplus 150) + 142 = 100 + 142 = 242$

Table 1 Hyper parameters of DNN

Parameter	Specifications
Learning Algorithm	Adam
Initial Learning Rate	0.001
Epoch	200
Iterations	600
Decay Rate	0.5
Loss Function	Cross-entropy

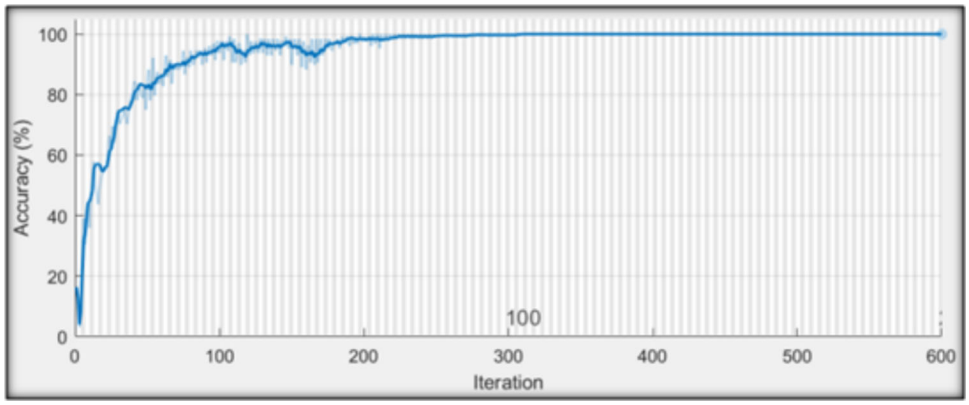


Fig. 5 Training accuracy (DNN-Adam)

Fig. 6 Training loss (DNN-Adam)

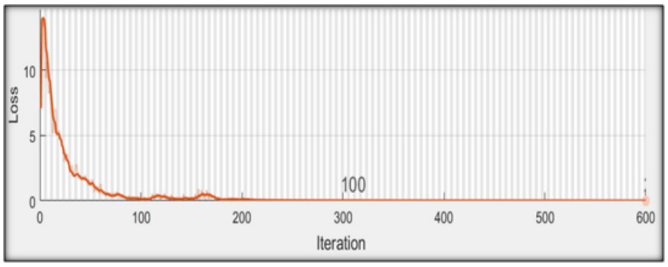


Table 2 DNN configuration

Layer	Hidden Layer	Output Activations	Total Learnable Parameters
Input Layer	-	10000×1	-
FC Layer-1	20	20×1	200020
ReLU-1	-	20×1	
BN-1	-	20×1	40
FC Layer-2	-	40×1	840
ReLU-2	40	40×1	
BN-2	-	40×1	80
FC Layer-3	60	60×1	1460
ReLU-3	-	60×1	
BN-3	-	60×1	120
FC layer-4	10000	10000×1	610000
Regression Layer		10000×1	

- **Step 7:** Convert the 1-D vector into a 2-D image as the encrypted image. Perform the reverse operation to get the decrypted image:

$$\mathrm{HSI}_{\mathrm{enc}} = \begin{bmatrix} 140 & 18 & 198 & 52 \\ 240 & 98 & 228 & 112 \\ 124 & 118 & 32 & 142 \\ 214 & 178 & 42 & 242 \end{bmatrix}$$

Experimental results and discussions

The proposed system is implemented on MATLAB using image processing and Deep Learning Toolbox on the personal computer system having 8GB RAM, core i5 processor, and Windows operating system. The outcomes of the DLHCIE scheme are validated on the five HSIs. Three images are considered for simulation: Pavia University (PU), Kennedy Space Center (KSC), IndianPines, Salinas, and Cuprite (CP), having more than 200 spectral bands. The hyperparameters considered for the DNN are summarized in the Table 1. The training accuracy and loss of the DNN for the Adam optimization algorithm for initial learning rate of 0.001 and 200 epoch are illustrated in Figs. 5 and 6 respectively. The DNN provides training accuracy of 100% for 100 epochs however the training process is continued till 200 epochs to ensure the stability of training of DNN. The DNN framework utilized for the key enhancements needs 813.3K trainable parameters.

The configurations of the DNN are presented in Table 2. It provides a total trainable parameters of 812560 and training time of 346 sec.

The outcomes of the DLHCIE scheme are validated on the five HSIs. Three images are considered for simulation: Pavia University (PU), Kennedy Space Center (KSC), IndianPines, Salinas, and Cuprite (CP), having more than 200 spectral bands.

The visualization of the encryption process for the KSC and IndianPines dataset are given in Figs. 7 and 8 respectively. It shows the selected HIS bands based on maximum entropy and SD, encrypted image using DLHCEA and decrypted images.

Table 3 presents a comparison between two image encryption algorithms. For all datasets, the DLHCEA outperforms the traditional HCEA. The MSE values are lower for

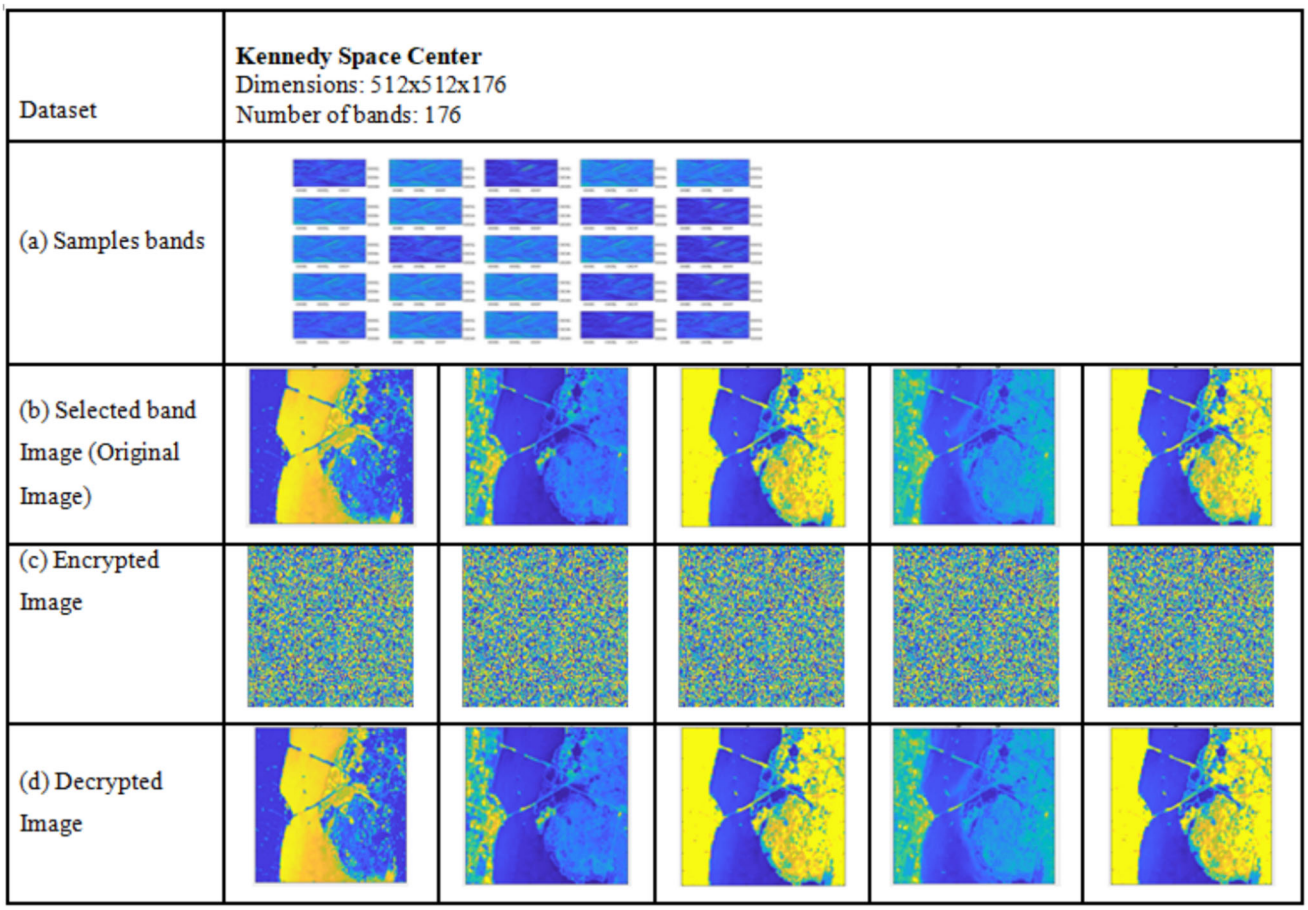


Fig. 7 Visualization of the DLHCE for KSC dataset a) Samples bands b) Selected band Image (Original Image) c) Encrypted Image d) Decrypted Image

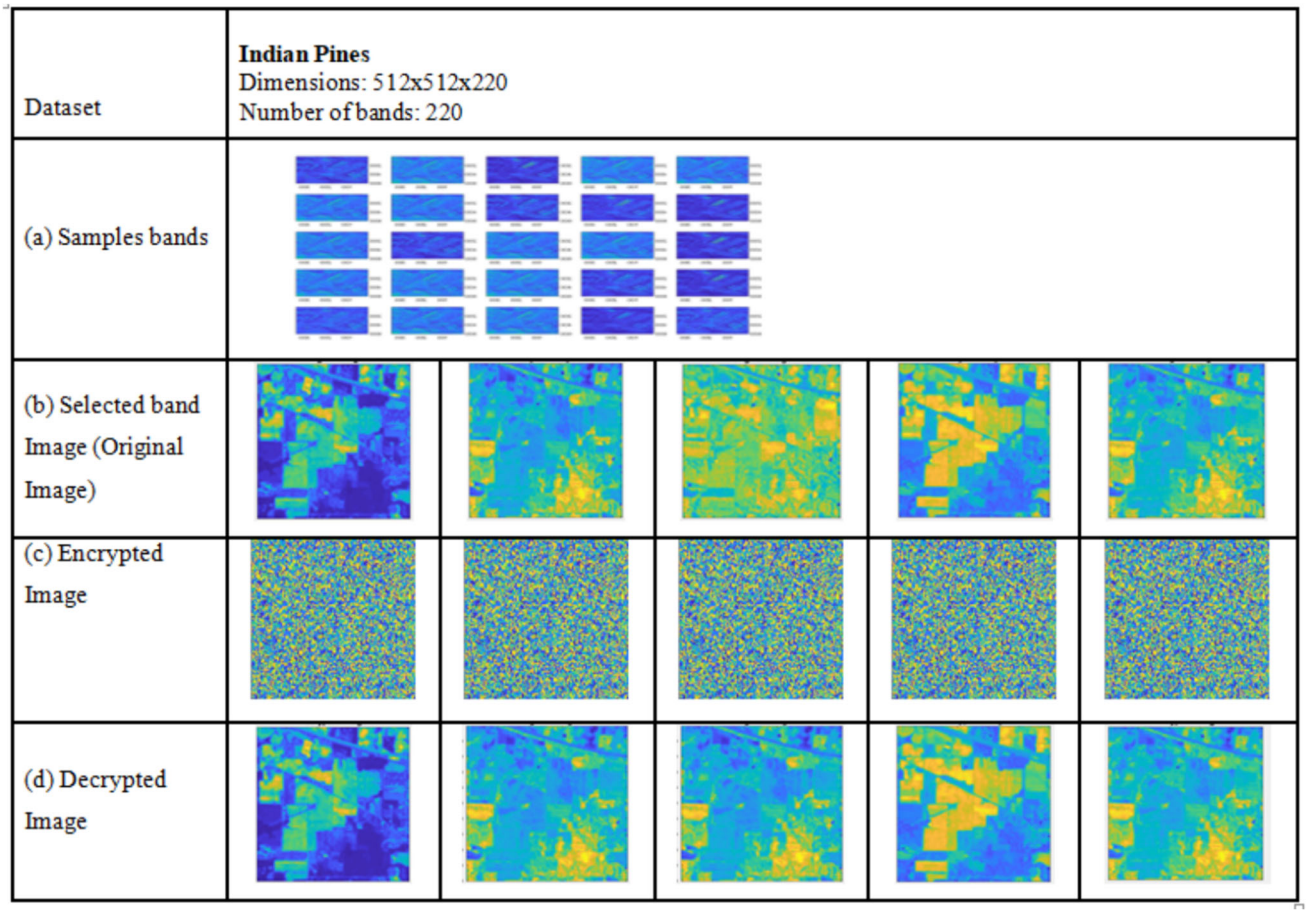


Fig. 8 Visualization of the DLHCE for Indian Pines dataset a) Samples bands b) Selected band Image (Original Image) c) Encrypted Image d) Decrypted Image

DLHCEA, indicating less error in the encrypted images. Correspondingly, the PSNR values are higher for DLHCEA, reflecting better reconstruction quality. Finally, the SSIM values show that DLHCEA maintains better structural similarity between the original and encrypted images compared to HCEA. For instance, for the KSC dataset, the DLHCEA achieves MSE of 1.946, PSNR of 45.23938, and SSIM of 0.974, outperforming the HCEA, which has an MSE of 2.72, a PSNR of 43.78511, and an SSIM of 0.956. The results suggest that incorporating deep learning into the hyper-chaotic encryption process leads to more effective encryption, with higher image quality and structural fidelity.

The DLHCEA offered a CC value closer to zero, indicating the statistical similarity between the original and encrypted bands. It offered the CC of 0.012 for KSC, 0.015 for Indian Pines, 0.011 for Salinas, 0.012 for PaviaU, and

Table 3 MSE, PSNR,SSIM,CC,NPCR and UACI for HCEA and DLHCE

	Hyper-chaotic Encryption Algorithm (HCEA)						(Deep Learning based Hyper-chaotic Encryption Algorithm) DLHCEA)					
Image	MSE	PSNR	SSIM	CC	NPCR (%)	UACI (%)	MSE	PSNR	SSIM	CC	NPCR (%)	UACI (%)
KSC	2.72	43.78511	0.956	0.015	98.12	26.54	1.946	45.23938	0.974	0.012	99.12	28.25
Indian Pines	2.84	43.59762	0.952	0.02	98.23	25.46	2.007	45.10533	0.961	0.015	99.24	27.12
Salinas	2.33	44.45724	0.976	0.016	97.89	27.59	1.593	46.10865	0.953	0.011	99.51	28.50
PaviaU	2.36	44.40168	0.961	0.017	98.17	27.60	1.52	46.31237	0.982	0.012	99.25	29.65
Cuprite	2.27	44.57055	0.979	0.025	98.34	26.85	1.54	46.2556	0.967	0.015	99.48	28.84

Table 4 Entrophy value for HCEA and DLHCE

Image	HCEA			DL-HCEA		
	Original	Encrypted	Decrypted	Original	Encrypted	Decrypted
KSC	7.245	7.839	7.246	7.245	7.923	7.467
Indian Pines	7.424	7.87	7.483	7.424	7.94	7.589
Salinas	7.098	7.882	7.259	7.098	7.945	7.545
PaviaU	7.409	7.832	7.175	7.409	7.903	7.562
Cuprite	7.356	7.818	7.258	7.356	7.897	7.41

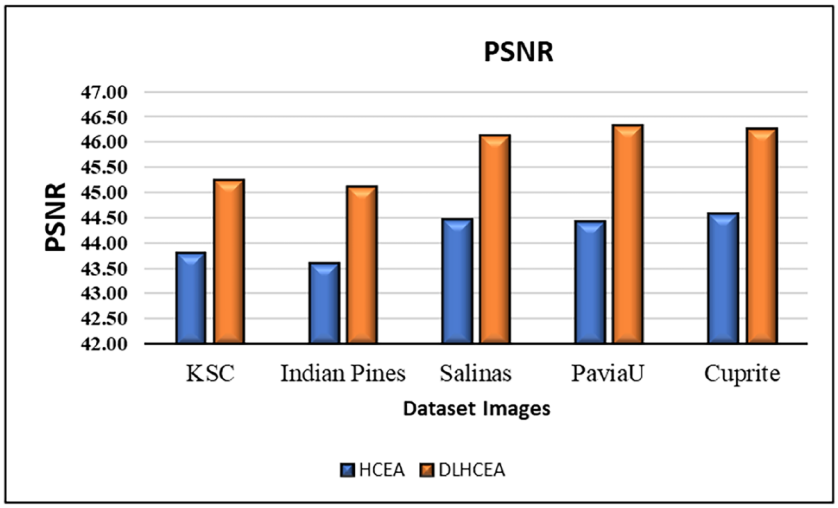


Fig. 9 Comparsion of PSNR for different images for HCEA and DLHCE

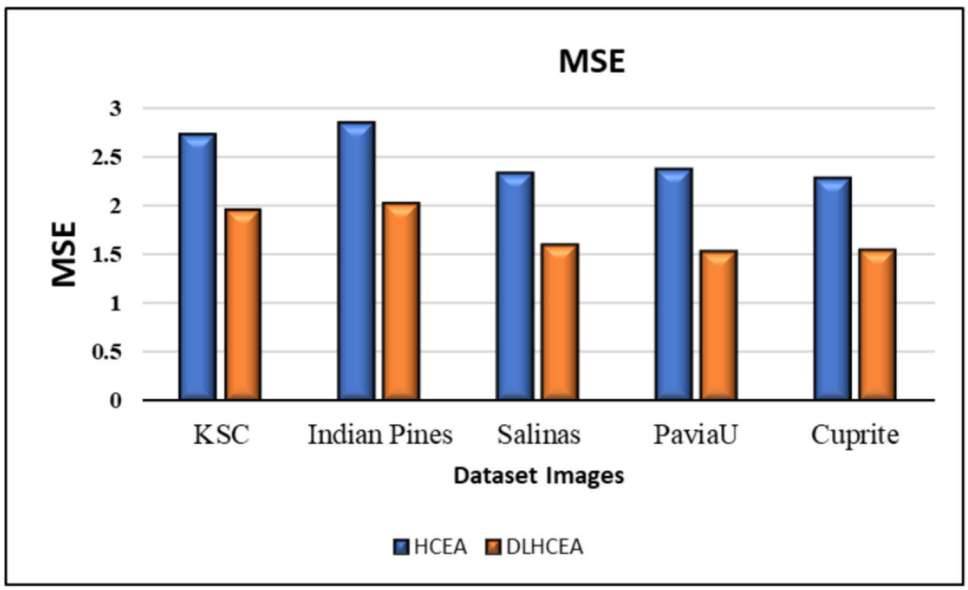


Fig. 10 Comparsion of MSE for different images for HCEA and DLHCE

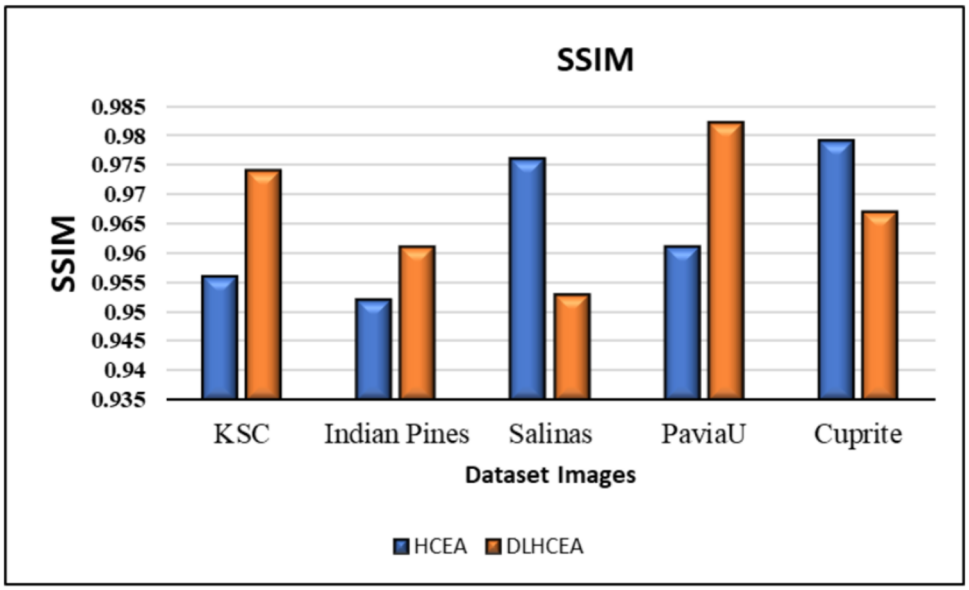


Fig. 11 Comparsion of SSIM for different images for HCEA and DLHCE

0.015 for Cuprite, which has shown significant improvement over the HCEA. The proposed DLHCEA offered the improved average NPCR of 99.12%, 99.24%, 99.51%, 99.25%, and 99.48% for the KSC, Indian Pines, Salinas, PaviaU, and Cuprite images. Considering the selective encryption bands helps maintain the structural similarity of the HSIs and offers a better NPCR. The UACI offers the average intensity disparity in the encrypted image. The DLHCEA provided enhanced UACI of 28.23% for KSC, 27.12% for Indian Pines, 28.50% for Salinas, 29.65% for PaviaU, and 28.84% for Cuprite, respectively.

Table 4 offers the entropy of the HCEA and DLHCEA. In the context of images, entropy measures the amount of information or randomness in the pixel values of the image. Higher entropy typically indicates more randomness or complexity, while lower entropy suggests a more structured image. In encryption, higher entropy of the encrypted image means that the encryption algorithm successfully increased the randomness of the data, making it harder to deduce the original information. For decrypted images, the entropy should be similar to the original if the decryption is successful.

The HCEA offers the entropy of 7.246, 7.483, 7.259, 7.175 and 7.258 for the KSC, Indian Pines, Salinas, PaviaU, and Cuprite dataset for decrypted images. Whereas, DLK-HCEA provides improved entropy for decryption than HCEA like 7.467 for KSC, 7.589 for Indian Pines, 7.545 for Salinas,

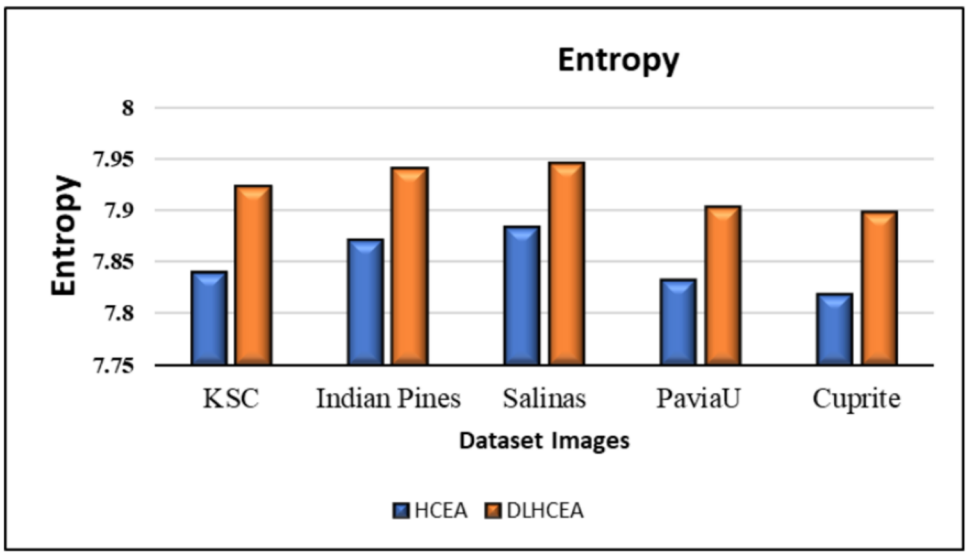


Fig. 12 Comparsion of Entrophy for different images for HCEA and DLHCE

7.562 for PaviaU and 7.41 for Cuprite. Both HCEA and DL-HCEA increase the entropy of the images when encrypted, which is expected. The encryption process should make the image appear more random (higher entropy). For all images, the entropy increases significantly in the encrypted state. DL-HCEA consistently has slightly higher encrypted entropy values compared to HCEA, suggesting that the DL-HCEA introduces slightly more randomness or complexity during encryption. In some cases (e.g., KSC, Indian Pines, Salinas), the DL-HCEA provides slightly better recovery of entropy in the decrypted image compared to HCEA, as the decrypted entropy values for DL-HCEA are closer to the original.

Figures 9, 10, 11 and 12 illustrate the comparison of the MSE, PSNR, SSIM and entrophy for the DLHCE and HCEA, respectively. The suggested DLHCE helps to retain the perceptual, visual, and structural quality of the HSIs compared with HCEA.

The histogram of the original and decrypted images shows the retention of the texture and structural content of the spectral bands, as shown in Figs. 13 and 14. The histogram of the encrypted images denotes complete randomness in the image content.

The results of the different alpha values for KSC dataset for five band selections are evaluated using PSNR and MSE, as given in Table 5. It is observed that α = 0.6 provides a higher weight to the entropy of the images. Higher entropy helps to select more diverse spectral bands and provides better visual quality and statistical results compared with lower values of the α. The α=0.6 provides balance in entropy and standard deviation, thus providing superior value for the band selection.

Analysis of attacks on encryption

The proposed DLHCEA image encryption method demonstrates robust resistance to three types of attacks: brute force attack (BFA), man-in-the-middle (MIM), and chosen-plaintext attacks (CPA). For brute force attacks, the hyperchaotic system's large key space, due to its high dimensionality and extreme sensitivity to initial conditions, makes it computationally infeasible to guess the correct key, as the complexity grows exponentially. In man-in-the-middle attacks, the encrypted communication remains secure due to the non-deterministic nature of the hyperchaotic maps, which makes it nearly impossible for an attacker to intercept and alter the communication without detection. A larger key space is essential for minimizing brute-force attacks. The hyperchaotic key has 256 bits in key length, providing a key space of 2^{256}. The larger key space makes it difficult to try every possible combination of the keys. A larger key space

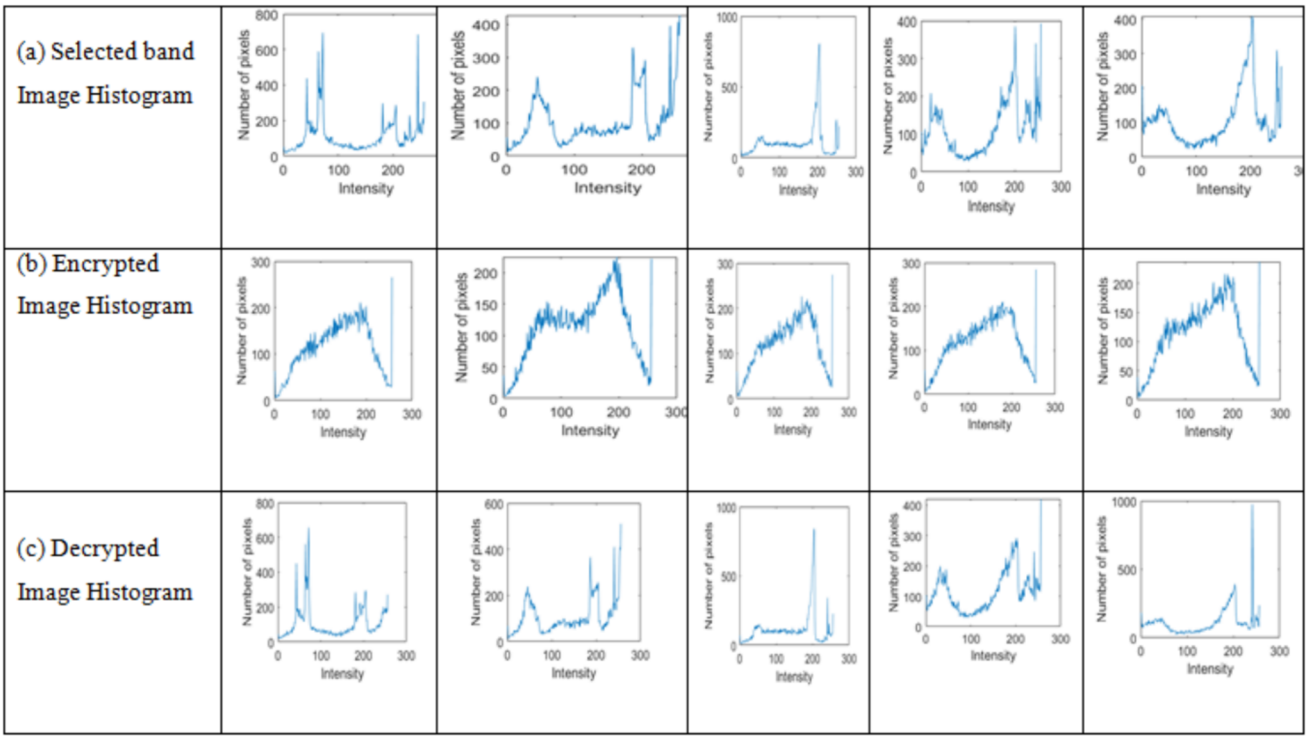


Fig. 13 Visualization of Histogram Analysis for KSC dataset (a)Selected band Image Histogram (b) Encrypted Image Histogram (c) Decrypted Image Histogram

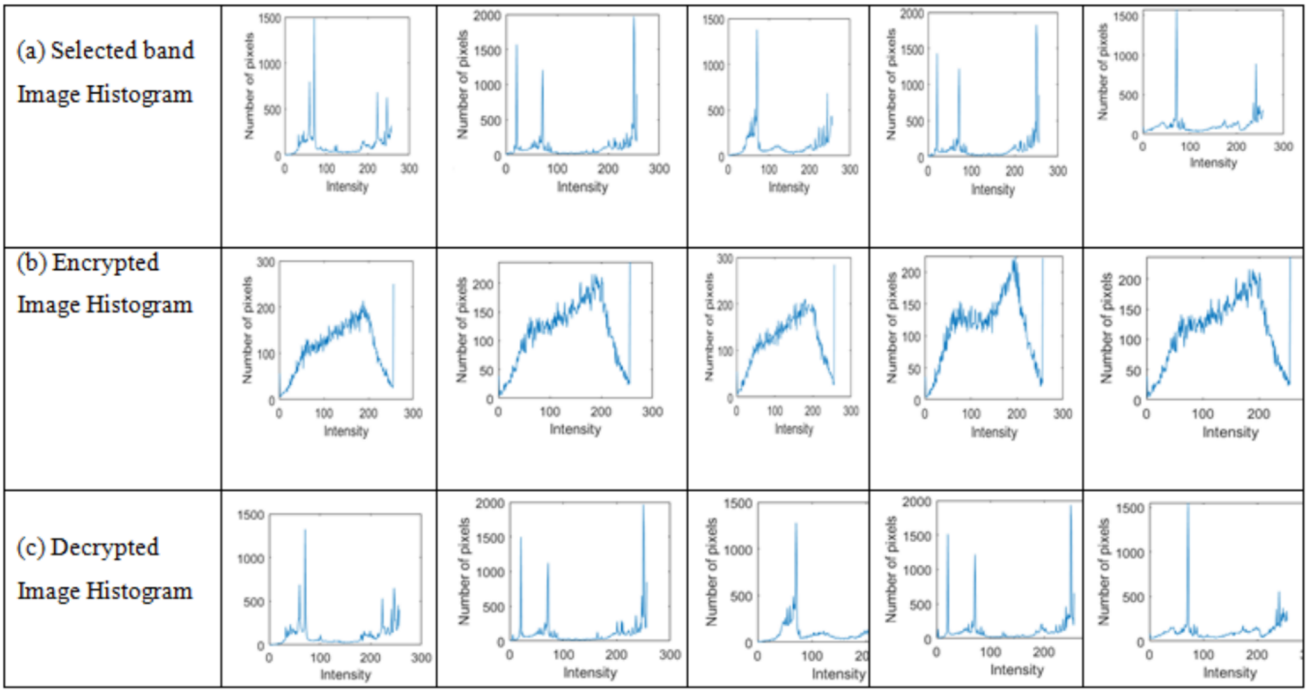


Fig. 14 Visualization of Histogram Analysis for Indian Pines dataset (a)Selected band Image Histogram (b) Encrypted Image Histogram (c) Decrypted Image Histogram

can accommodate more complex encryption techniques and most suitable for HSI data.

For chosen-plaintext attacks, the deep learning component enhances security by learning complex non-linear relationships between plaintext and ciphertext, ensuring that even if an attacker has access to chosen inputs and their corresponding encrypted outputs, reverse-engineering the encryption process is highly improbable. These results underscore the method's superior ability to safeguard hyperspectral images from common cryptographic attacks. The samples of the image encryption attack and its decrypted HIS are given in Fig. 15. The DL-HCEA enhances key sequence and hence the attacks fails to decrypt the image in its original format. The decrypted images using distinct attacks shows the higher entropy of 7.85, 7.86, 7.83 for the BFA, MIM and CPA based attacks which is greater than entropy of original band.

Conclusion

Hyperspectral image encryption is crucial for maintaining the security, authenticity, and privacy of the intricate multi-

Table 5 Evaluation of α value for KSC dataset

Value of α	MSE	PSNR	SSIM	CC	NPCR (%)	UACI (%)
α =0.1	36.246	32.538	0.924	0.072	76.60	17.20
α =0.2	23.161	34.483	0.934	0.067	82.25	21.37
α =0.3	17.082	35.805	0.956	0.046	92.24	23.87
α =0.4	13.553	36.810	0.975	0.035	97.45	25.76
α =0.5	3.363	42.863	0.967	0.024	98.65	26.00
α =0.6	**1.946**	**45.239**	**0.974**	**0.012**	**99.12**	**28.25**
α=0.7	4.136	41.965	0.965	0.018	98.20	27.45
α =0.8	12.456	37.177	0.953	0.025	97.45	26.34
α =0.9	23.765	34.371	0.942	0.045	95.36	24.86

Entries in bold presents the proposed method value for $\alpha = 0.6$ gives better result as compared to other value

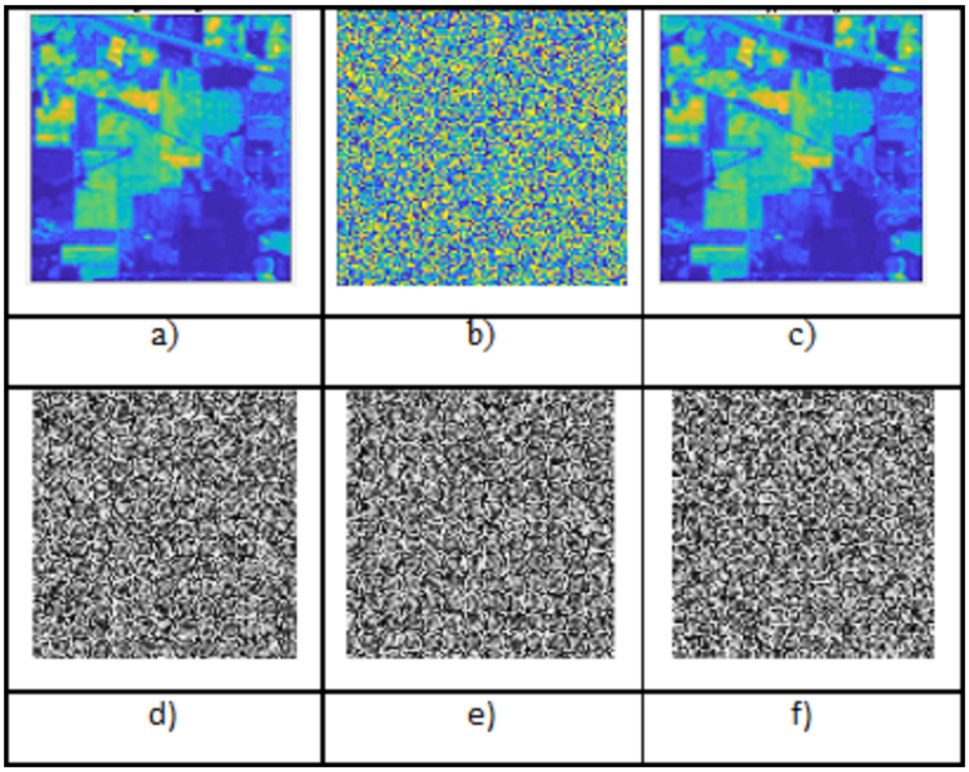


Fig. 15 Visualization of attack performance a) original Image b) Selected band c) Encrypted band using DLHCEA d) Decrypted by using DLHCEA e) Decrypted image after BFA f) Decrypted image after MIM g) Decrypted image after CPA

band images. This paper presents a DNN-based hyperchaotic encryption algorithm to enhance the security and privacy of hyperspectral images. The suggested algorithm uses DNN to optimize the randomness of the encryption key generated during the hyperchaotic encryption process. The DLHCE offers the robustness and reliability for the HSIE using selective band encryption techniques based on entropy and variance. The DLHCE provides the MSE in the range of 1.946 to 2.007, PSNR in the range of 45.10 dB to 46.31dB, and SSIM in the range of 0.953 to 0.982 for five sample high-dimensional HSIs. The DLHCE offers improved security, reliability, and robustness for encryption attacks such as BFA, MIM, and CPA. It achieves a better entropy of 7.85 for a brute force attack, 7.86 for a man-in-middle attack, and 7.83 for the chosen plaintext attack. The hyper-chaotic key sequence's higher randomness offers a faster, more efficient way of image encryption. The encryption algorithm shows robustness to various attacks because the randomness of the hyper-chaotic encryption sequence has been optimized. In the future, the system's performance can be enhanced by inculcating the DL algorithm for encrypting HSIs. The system's effectiveness can also be enhanced by optimizing the hyper-parameters of the chaotic encryption algorithm. In the future, the watermarking technique can be utilized to hide the authentication mark inside the spectral bands for HIS authentication.

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Declarations

Conflicts of Interest The authors declare no conflict of interest.

Ethical Approval Not Applicable.

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